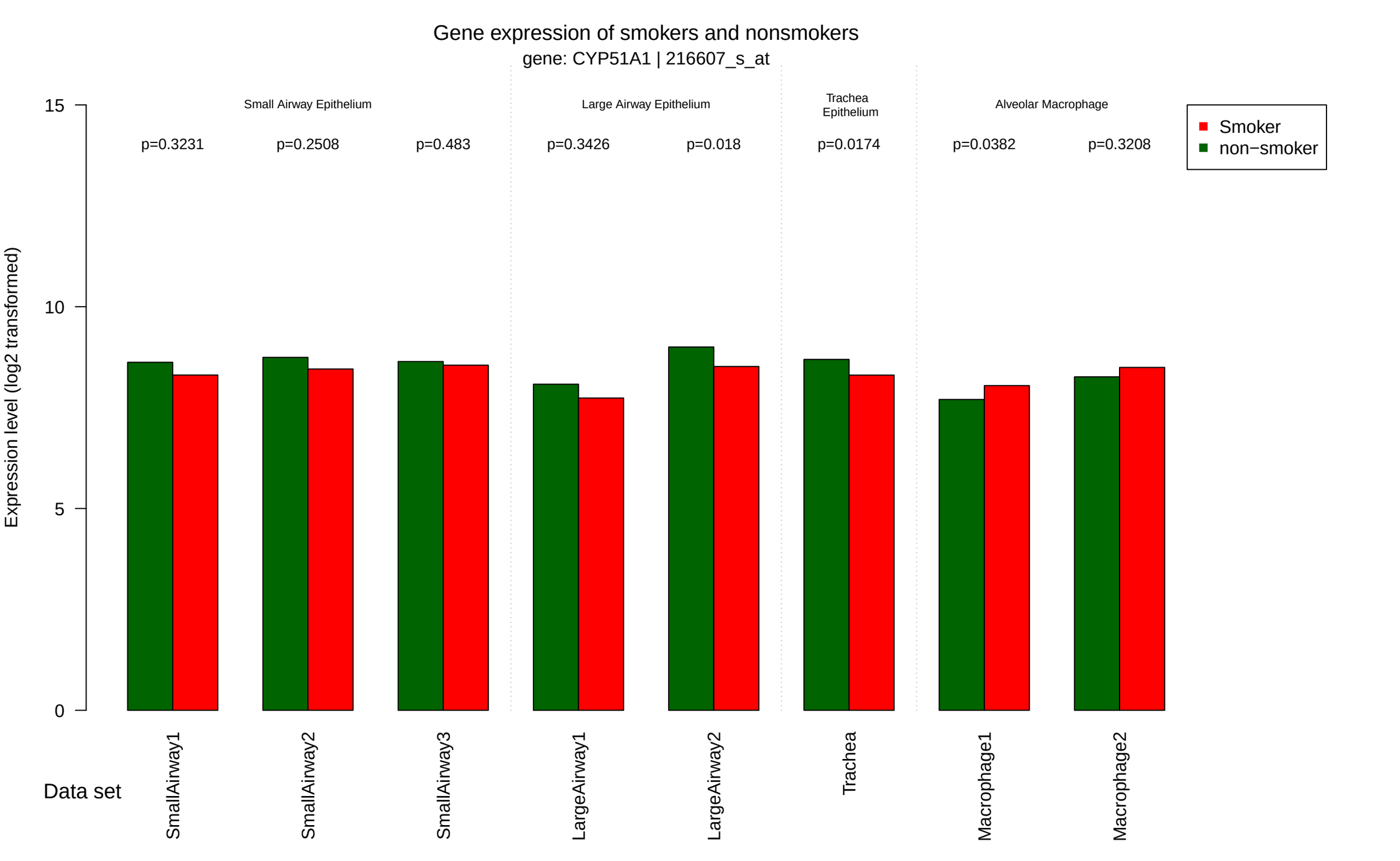
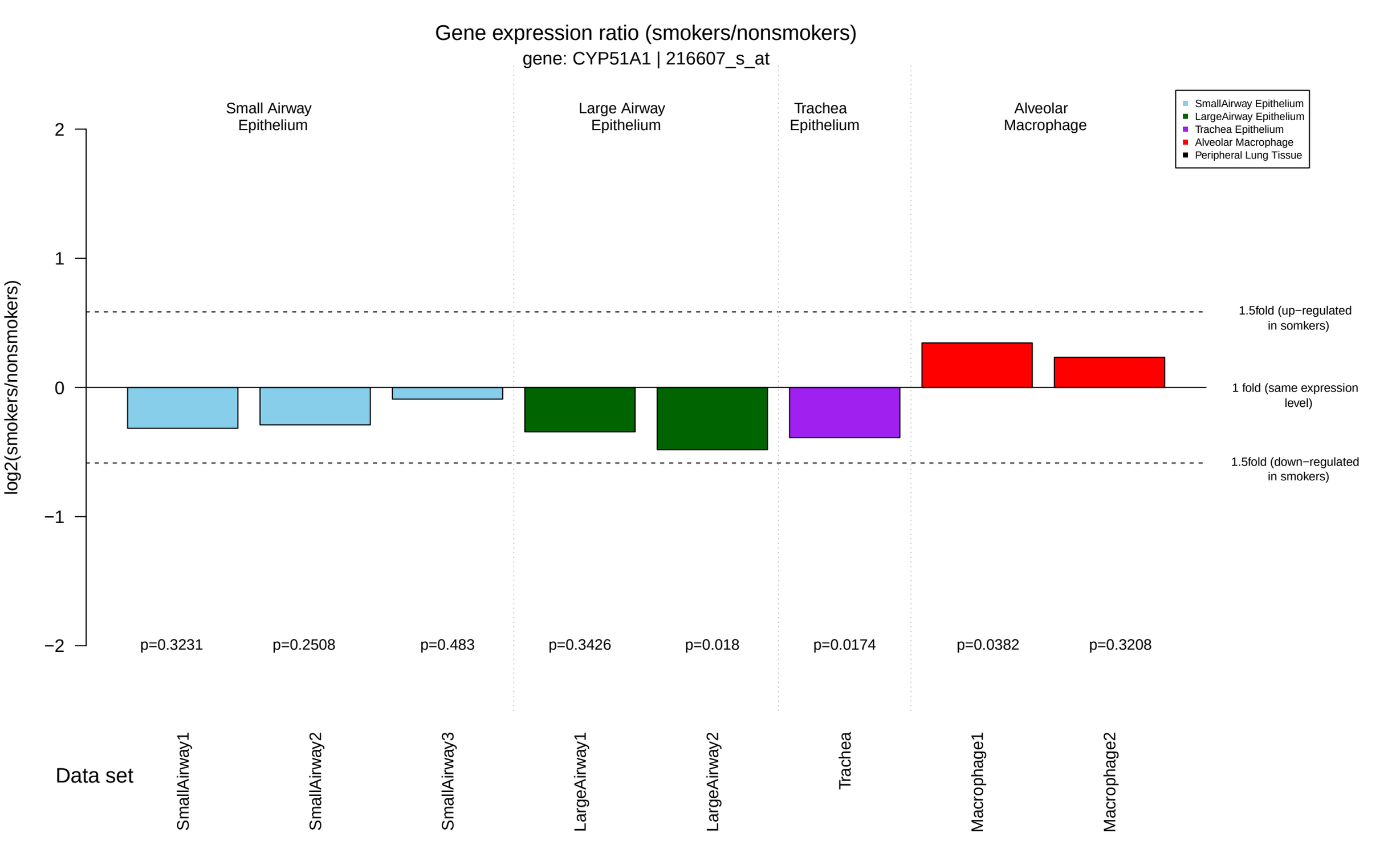
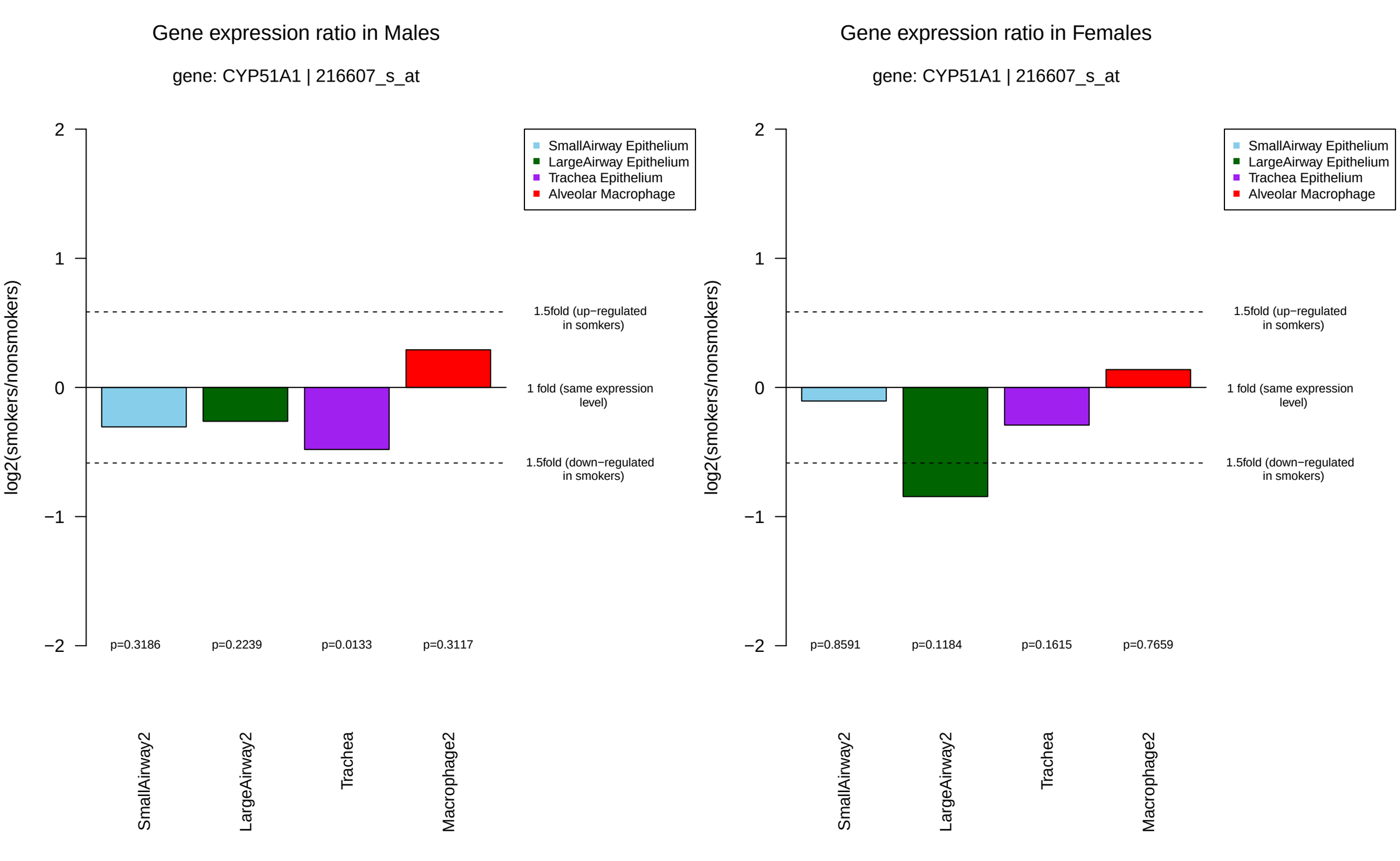
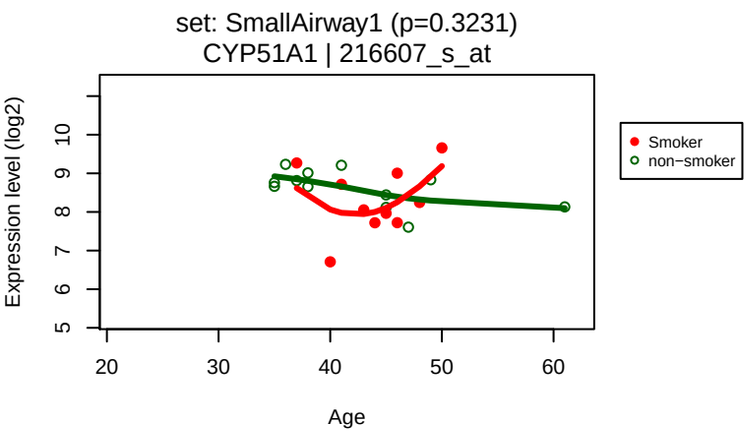
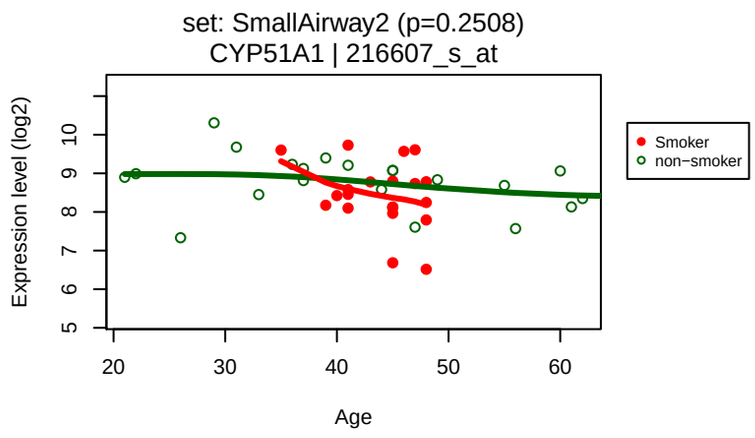
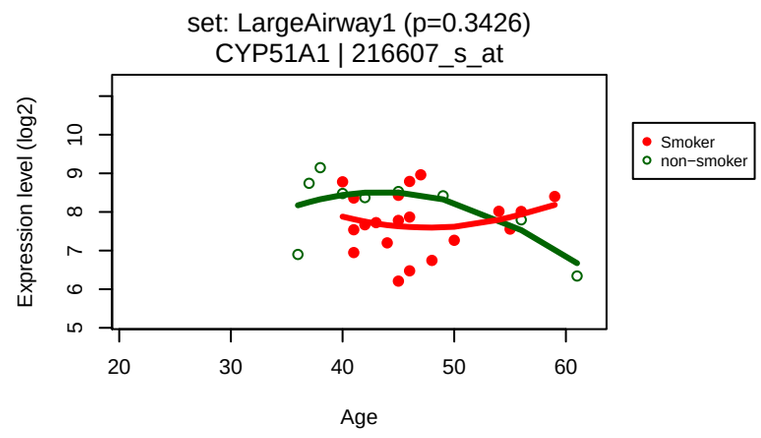

Gene expression of smokers and nonsmokers
gene: CYP51A1 | 216607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3231
p=0.2508
p=0.483
p=0.3426
p=0.018
p=0.0174
p=0.0382
p=0.3208
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

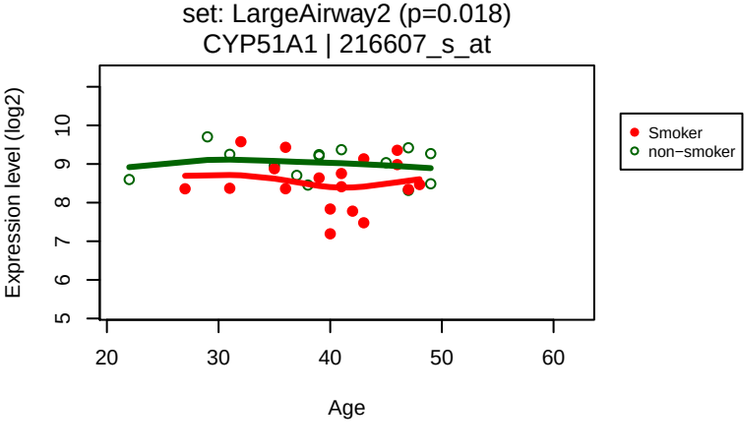

Gene expression ratio (smokers/nonsmokers)
gene: CYP51A1 | 216607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3231
p=0.2508
p=0.483
p=0.3426
p=0.018
p=0.0174
p=0.0382
p=0.3208
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

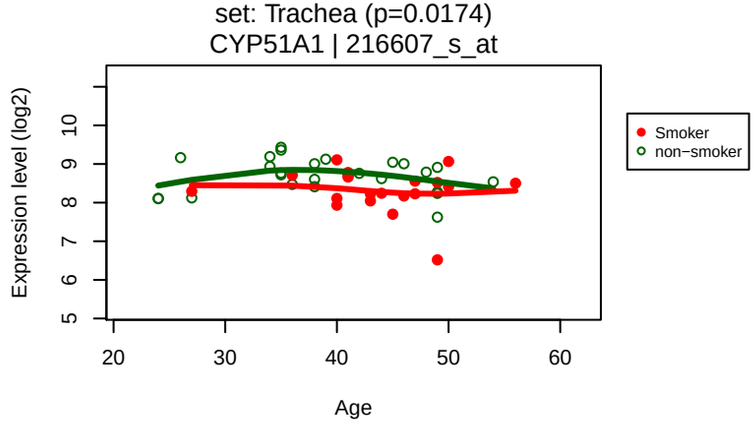

Gene expression ratio in Males
gene: CYP51A1 | 216607_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3186
p=0.2239
p=0.0133
p=0.3117
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP51A1 | 216607_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8591
p=0.1184
p=0.1615
p=0.7659
SmallAirway2
LargeAirway2
Trachea
Macrophage2

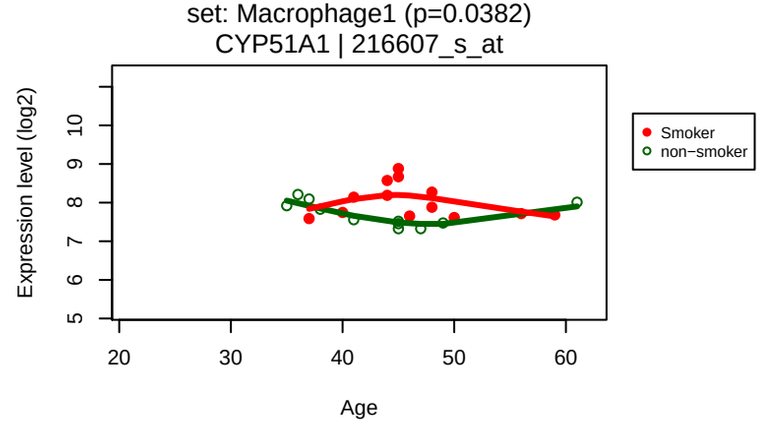

set: SmallAirway1 (p=0.3231)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker

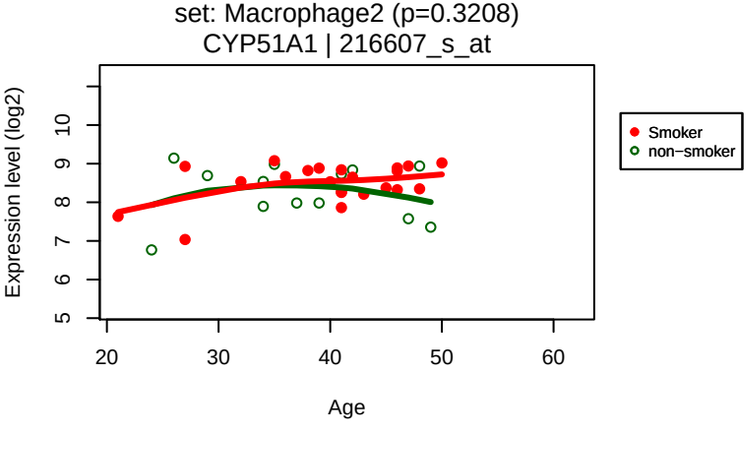

set: SmallAirway2 (p=0.2508)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker

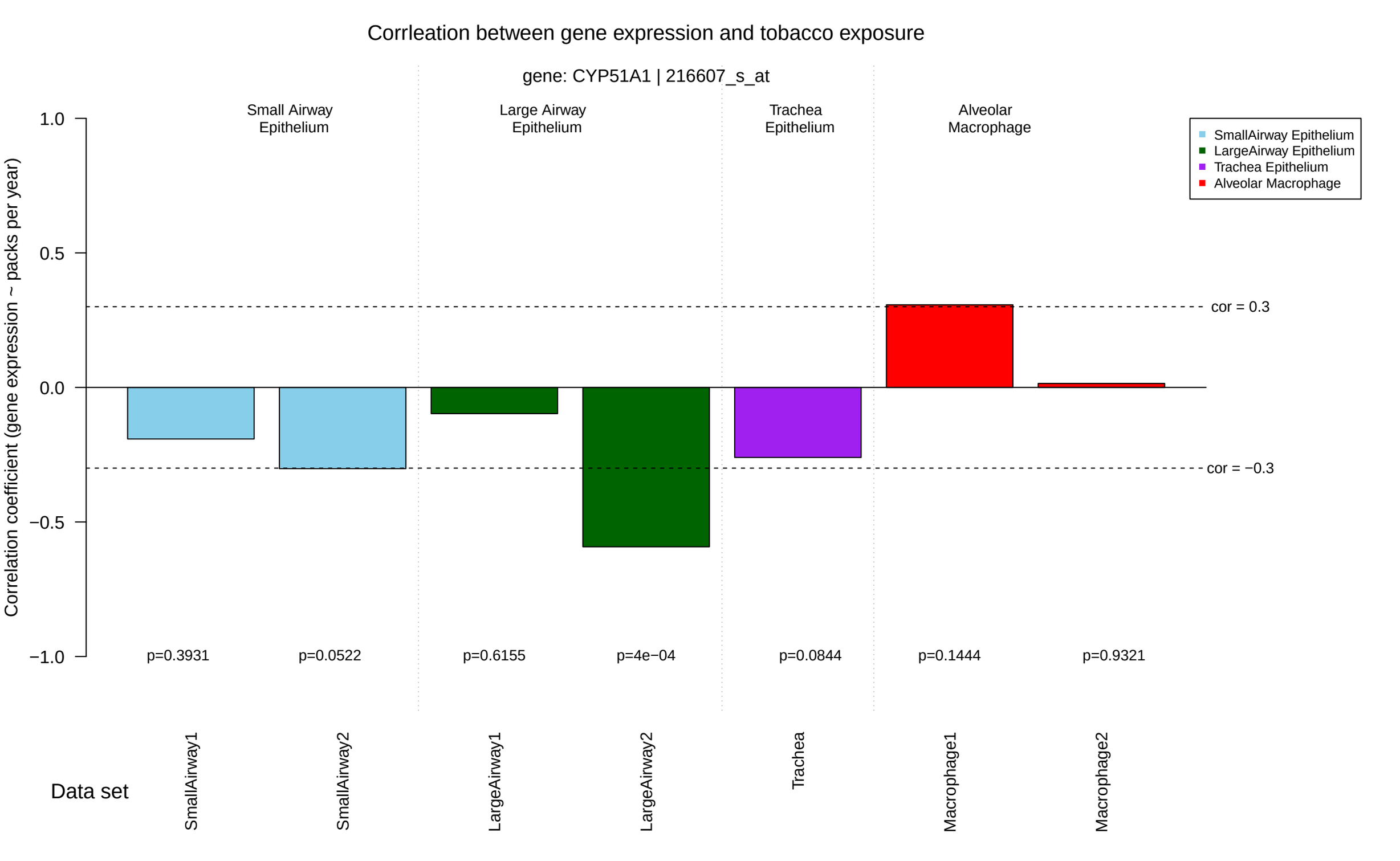

set: LargeAirway1 (p=0.3426)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.018)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0174)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0382)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3208)
CYP51A1 | 216607_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP51A1 | 216607_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3931
p=0.0522
p=0.6155
p=4e−04
p=0.0844
p=0.1444
p=0.9321
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2